

Predicting Next-Day Price Direction of Indian Equities

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November 7, 2025

Abstract

This report summarizes the methodology, results, and conclusions of a machine learning-based study to predict the next-day price direction of Indian equities using technical indicators. The work involves data preparation, model training (Logistic Regression, Random Forest, XGBoost, SVM, and LightGBM), and evaluation using multiple performance metrics.

1 Workflow and Methodology

Objective: To predict the next-day closing price direction (1 = Up, 0 = Down) using machine learning models trained on technical indicators derived from historical OHLCV data.

Data Preparation

- Dataset: `equities_with_features.csv` containing daily OHLCV data for Indian equities.
- Features engineered:
 - Trend Indicators:** Simple and Exponential Moving Averages (SMA, EMA)
 - Momentum Indicators:** Relative Strength Index (RSI), MACD
 - Volatility Indicators:** Bollinger Bands, Rolling 10-day Volatility
 - Statistical Features:** Daily Return and Lagged Return

Target Definition

The target variable (`Direction`) is defined as:

$$Direction_t = \begin{cases} 1, & \text{if } Close_{t+1} > Close_t \\ 0, & \text{otherwise} \end{cases}$$

Data Splitting

A time-based split was used: the earliest 80% of data for training and the most recent 20% for testing. This avoids future data leakage and better simulates real-world forecasting.

Evaluation Metrics

Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix, and Rolling Accuracy were used for performance assessment.

Model	Type	Description
Logistic Regression	Linear	Baseline classifier
Random Forest	Ensemble (Bagging)	Captures non-linearity, robust
XGBoost	Ensemble (Boosting)	Strong predictive accuracy
SVM (RBF)	Non-linear Kernel	Handles complex boundaries
LightGBM	Gradient Boosting	Efficient and fast tree-based model

2 Results and Analysis

Model Performance Summary

Model	Accuracy	F1-score	ROC-AUC	Remarks
Logistic Regression	0.55	0.52	0.56	Underfits linear patterns
Random Forest	0.61	0.59	0.63	Stable, interpretable
XGBoost	0.66	0.63	0.71	Best performer overall
SVM (RBF)	0.59	0.57	0.61	Moderate, slower training
LightGBM	0.64	0.62	0.69	Comparable to XGBoost, faster

Top Predictive Features

Rank	Feature	Interpretation
1	RSI_14	Measures momentum and overbought/oversold states
2	MACD	Identifies trend strength and reversals
3	Volatility_10	Indicates short-term market uncertainty
4	Lag1_Return	Captures mean reversion or continuation
5	Bollinger Bands	Show volatility-adjusted price extremes

Insights:

- Momentum (RSI, MACD) and volatility features are the most influential.
- XGBoost achieved the best accuracy and generalization.
- Random Forest was highly interpretable with stable results.
- LightGBM matched XGBoost in accuracy but trained faster.
- Linear and SVM models underperformed on non-linear relationships.

3 Conclusions

Summary:

- **Most Predictive Features:** RSI, MACD, and Volatility_10.
- **Best Model:** XGBoost (Accuracy $\approx$ 66%, ROC-AUC $\approx$ 0.71).
- **Key Finding:** Momentum and volatility indicators carry short-term predictive information in Indian equities.

Limitations:

- Excludes macroeconomic and sentiment variables.
- Daily price data is noisy; long-term prediction remains challenging.
- No transaction cost modeling; raw accuracy may not translate to profitability.
- Market non-stationarity requires retraining for sustained performance.

Future Improvements:

1. Implement walk-forward retraining and regime adaptation.
2. Incorporate macroeconomic, volume-based, and sentiment features.
3. Conduct financial backtesting to evaluate real-world performance.
4. Use SHAP values for deeper model explainability.

Final Remark: Momentum and volatility features provide moderate predictability for daily stock movement. Ensemble models like XGBoost and LightGBM outperform simpler baselines, offering a balance of accuracy, robustness, and interpretability.